Neuroanatomy Deep Gray Nuclei

🧩 Neuroanatomy — Deep Gray Nuclei & Basal Ganglia

> *The basal ganglia form a central switching station for motor, cognitive, and emotional programs — shaping everything from buttoning a shirt to suppressing an urge.* — Adams & Victor

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🎯 Core Concept

Deep gray nuclei are collections of neuronal cell bodies buried within the cerebrum, brainstem, and cerebellum. The **basal ganglia** are the most clinically significant group — a set of interconnected nuclei that act as the brain's **action selection and modulation** system.

🧩 Deep Gray Nuclei — Complete Map

Nucleus	Location	Primary Function
Caudate nucleus	Lateral ventricle wall (head, body, tail)	Motor planning, cognitive sequencing
Putamen	External capsule ⬚ insular cortex	Motor execution, habit formation
Globus pallidus externus (GPe)	Between putamen and GPi	Inhibitory regulator of STN
Globus pallidus internus (GPi)	Medial to GPi, lateral to internal capsule	DBS target; main basal ganglia output
Subthalamic nucleus (STN)	Midbrain, rostral to substantia nigra	Excitatory regulator of GPi
Substantia nigra pars reticulata (SNr)	Midbrain, medial to cerebral peduncle	Output to thalamus; movement gating

Nucleus	Location	Primary Function
Substantia nigra pars compacta (SNc)	Midbrain, dorsal to SNr	Dopamine production → movement facilitation
Thalamus	Lateral ventricles, medial to internal capsule	Relay station (sensorimotor/ cognitive)
Amygdala	Temporal pole, deep to uncus	Fear, emotion, memory modulation
Hippocampus	Medial temporal lobe, in hippocampal formation	Episodic memory, spatial navigation

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🔗 Basal Ganglia Circuitry

The Motor Loop (Most Studied)

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CORTEX (Motor/Sensorimotor)
        ↓ (Excitatory: Glutamate)
STRIA TUM (Caudate + Putamen)
        ↓
    ┌──────────────────────────┐
    │   DIRECT PATHWAY         │     INDIRECT PATHWAY
    │   (Facilitates movement)│     (Suppresses movement)
    │   Striatum → GPi/SNr     │     Striatum → GPe → STN → GPi/SNr
    │   (Inhibits, ↓GABA)      │     (Disinhibits, ↑Excitatory)
    └──────────────────────────┘
        ↓                              ↓
        ↓                              ↓
   GPi/SNr ──────── THALAMUS ──────── GPi/SNr
   (Inhibits, ↓GABA)      ↑                  (Inhibits more,
↓↓GABA)
        ↓                              ↓
        ↓                              ↓
   EXCITE CORTEX                  EXCITE CORTEX (less)
   (More movement)                (Less movement)
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Direct Pathway (Go)

1. Cortex excites Striatum (glutamate)
2. Striatum inhibits GPi/SNr (GABA)
3. GPi/SNr releases thalamus → Cortex gets **more excitation**
4. Result: **Movement is facilitated**

Indirect Pathway (Stop)

1. Cortex excites Striatum

2. Striatum inhibits GPe (GABA)
3. GPe releases STN → STN **more active**
4. STN excites GPi/SNr (glutamate)
5. GPi/SNr inhibits thalamus → Cortex gets **less excitation**
6. Result: **Movement is suppressed**

Dopamine Modulation

Receptor	Effect on pathways
D1 receptors (SNc → Striatum)	Excites **direct** pathway → facilitates movement
D2 receptors (SNc → Striatum)	Inhibits **indirect** pathway → suppresses less = more movement

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⚡ Clinical Correlations

Condition	Basal Ganglia Pathology	Key Features
Parkinson Disease	Loss of SNc dopaminergic neurons	Bradykinesia, rigidity, resting tremor, postural instability
Huntington Disease	Caudate nucleus atrophy	Chorea, psychiatric symptoms, cognitive decline
Hemiballismus	Contralateral STN lesion	Wild, flinging movements on one side
Dystonia	Basal ganglia dysfunction	Sustained or intermittent muscle contractions
Tic disorders (Tourette)	Cortico-striato-thalamic circuit dysfunction	Motor and vocal tics

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🧠 Parkinson Disease — Basal Ganglia Correlate

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Normal:
SNc → Dopamine → D1 (↑direct) + D2 (↓indirect)
      → GPi/SNr activity balanced
      → Thalamus properly gated → smooth movement

Parkinson:
SNc DEGENERATION → ↓Dopamine
      → GPi/SNr OVERACTIVE (too much inhibition on thalamus)
      → Thalamus UNDERACTIVE → Cortex gets too little excitation
      → Bradykinesia + Rigidity
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Treatment (DBS of GPi or STN):
      →人⍰ ↓GPi activity → restore thalamic excitation → improve
movement
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🎯 Board Pearls

> **What separates the caudate from the putamen?** The **internal capsule** (white matter tract of corticospinal fibers).

> **DBS target for Parkinson?** GPi (globus pallidus internus) or STN (subthalamic nucleus).

> **Hemiballismus** = lesion of the **contralateral STN** (lacunar stroke, classically)

> **Huntington disease** = CAG trinucleotide repeat; shows **caudate atrophy** → enlarged lateral ventricles (ex vacuo hydrocephalus)

> **Two major outputs of basal ganglia:** GPi/SNr → Thalamus (ventral anterior + ventrolateral nuclei) → Cortex

> **D1 receptors** are **excitatory** on direct pathway; **D2 receptors** are **inhibitory** on indirect pathway

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📊 Basal Ganglia Pathways Summary

Feature	Direct (Go)	Indirect (Stop)
Effect	Facilitates movement	Suppresses movement
Striatal output neurons	D1-receptor bearing	D2-receptor bearing
Path	Striatum → GPi/SNr → Thalamus	Striatum → GPe → STN → GPi/SNr → Thalamus
Dopamine effect	↑ (facilitates go)	↓ (reduces stop = more go)
Parkinsonism	Underactive	Overactive

📋 Clinical Case

A 62-year-old man presents with: - Resting tremor of right hand (pill-rolling) - Bradykinesia - Rigidity (cogwheel type) - Shuffling gait with reduced arm swing

Question: Where is the lesion? > **Substantia nigra pars compacta** — loss of dopaminergic neurons → decreased dopamine in striatum → overactive GPi/SNr → underactive thalamus → reduced cortical excitation → bradykinesia/tremor/rigidity

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📚 Sources

- **Bradley** — *Neurology in Clinical Practice* (8th ed)
- **Adams & Victor** — *Principles of Neurology* (12th ed)
- **Alberstone et al.** — *Anatomic Basis of Neurologic Diagnosis* (2nd ed)
- **Localization in Clinical Neurology** (8th ed) — Papa
- **Radiopaedia** — Basal ganglia pathways

🔗 Related Topics

- [[wiki/foundation/NEUROANATOMY_GROSS_BRAIN|Gross Brain — Lobes & Cortex]]
- [[wiki/foundation/NEUROANATOMY_CRANIAL_NERVES|Cranial Nerves Overview]]
- [[wiki/foundation/NEUROANATOMY_BLOOD_SUPPLY|Blood Supply — Cerebral Circulation]]

🃏 Anki Prompts

Front	Back
Direct pathway of basal ganglia — effect on movement?	Facilitates movement (Go)
Indirect pathway of basal ganglia — effect on movement?	Suppresses movement (Stop)
D1 receptor location and effect?	Striatum; excites direct pathway → facilitates movement
D2 receptor location and effect?	Striatum; inhibits indirect pathway → disinhibits movement
Parkinson disease — what degenerates?	Substantia nigra pars compacta
Hemiballismus — lesion of which structure?	Contralateral subthalamic nucleus
Huntington disease — what is seen on imaging?	Caudate nucleus atrophy, ex vacuo hydrocephalus
DBS target for Parkinson — two options?	GPi (globus pallidus internus) or STN (subthalamic nucleus)
What separates caudate from putamen?	Internal capsule
Parkinson — why is there too much GPi/SNr activity?	Loss of dopamine → disinhibition of indirect pathway → STN overactive → GPi overactive
SNr stands for?	Substantia nigra pars reticulata
Thalamus receives input from which basal ganglia structures?	GPi and SNr
Two major outputs of basal ganglia to thalamus?	GPi → VA/VL thalamus; SNr → VA/VL thalamus

Front	Back
Huntington disease — genetic mutation?	CAG trinucleotide repeat expansion in huntingtin gene
Huntington — triad?	Chorea + psychiatric symptoms + cognitive decline

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